

Timothy Stephens

Curriculum vitae

Personal summary

I am a computational biologist working on understanding and characterizing functionally uncharacterized “dark genes”, which are found in all non-model genomes, but often ignored despite their immense biological importance. I am also CTO of OceanOmics, a company dedicated to developing scalable tools for environmental health monitoring.

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Education and Research Experience

- 2024 — present Research Associate, Department of Biochemistry and Microbiology, Rutgers University
- 2019 — 2024 Post Doctoral Associate, Department of Biochemistry and Microbiology, Rutgers University
- 2016 — 2019 PhD, Institute for Molecular Bioscience, The University of Queensland
- 2015 — 2015 Honours, Institute for Molecular Bioscience, The University of Queensland
- 2012 — 2015 Bachelor of Biotechnology, Institute for Molecular Bioscience, The University of Queensland

Selected Awards and honours

- 2023 Selected to attend and present at the New Phytologist Next Generation Scientists Conference 2023, National University of Singapore
- 2021 2019 Dean’s Award for Outstanding Higher Degree by Research Theses, University of Queensland
- 2016 Research Training Program (RTP) scholarship, University of Queensland
- 2012 Dean’s Commendation for Academic Excellence (achieved a GPA over 6.6), University of Queensland
- 2011 Merit Scholarship for academic achievement, University of Queensland

Select Publications

- 2026 Nandi S., **Stephens T. G.**, Garcia R., Sanchez-Garcia M. A., Roberson L. M., Avalos J., Chundawat S. P. S., and Bhattacharya D. Rafts of Change: Microbial and Functional Dynamics in Simulated Sargassum Strandings. *Applied and Environmental Microbiology*, 92:e02357-25, 2026. [\[URL\]](#)
- 2026 **Stephens T. G.**, Kulczyk A. W, and Bhattacharya D. Cosmopolitan gene families with known functions are hotspots for the evolution of novel genes in stony corals. *GBE*, 18:evag072-4, 2026. [\[URL\]](#)
- 2026 *Nandia S., ***Stephens T. G.**, Chille E. E., Goyen S., Bay L. K., and Bhattacharya D. Metaproteome Analysis of Short-Term Thermal Stress in Three Sympatric Coral Species Reveals Divergent Host Responses. *Ecology and Evolution*, 16:e73275, 2026. [\[URL\]](#)
- 2026 Karimi K., Arroyo M., Chille E. E., **Stephens T. G.**, Barrett D., Pelechano V., Bhattacharya D., and Javanmard M. A portable, low-cost, point-of-care DNA amplification kit with impedance-based detection for decentralized antimicrobial resistance diagnostics. *Lab on a Chip*, 2026. [\[URL\]](#)

- 2026 Bhattacharya D., Nandi S., Chille E. E., Arroyo M., and **Stephens T. G.** The host coral bleaching response viewed through the lens of multi-omics. *BioEssays*, e70110, 2026. [[URL](#)]
- 2025 Nandi S., **Stephens T. G.**, Walsh K., Garcia R., Villalpando M. F., Sellares-Blasco R. I., Zubillaga A. L., Croquer A., Bhattacharya D. Shifts in the microbiome and virome are associated with stony coral tissue loss disease (SCTLD). *ISME Communications*, ycaf226, 2025. [[Preprint](#)] [[URL](#)]
- 2024 **Stephens T. G.**, Van Etten J., McDermott T., Christian W., Chaverra M., Gurney J., Lee Y., Kim H., Hyun Cho C., Chovancek E., Westhoff P., Otte A., Northen T. R., Bowen B. P., Louie K. B., Barry K., Grigoriev I. V., Mock T., Liu S., Miyagishima S., Yoshinaga M., Weber A. P.M., Yoon H. S., and Bhattacharya D. Temporal dynamics in a red alga dominated geothermal feature in Yellowstone National Park. *ISME Communications*, 4:01, 2024. [[Preprint](#)] [[URL](#)]
- 2024 Benites L. F., **Stephens T. G.**, Etten J. V., James T., Christian W. C., Barry K., Grigoriev I. V., McDermott T. R. & Bhattacharya D. Viruses associated with extremophilic red algal mats reveal signatures of early thermal adaptation. *Communications Biology*, 2024. Featured by [Science](#) [[URL](#)]
- 2022 *Gabr A., ***Stephens T. G.**, and Bhattacharya D. Loss of key endosymbiont genes may facilitate early host control of the chromatophore in *Paulinella*. *iScience*, 25:104974, 2022. *Co-first authorship [[URL](#)]
- 2022 *Calatrevra V., ***Stephens T. G.**, Gabr A., Grossman A. R., and Bhattacharya D. Retrotransposition facilitated the establishment of a primary plastid in the thecate amoeba *Paulinella*. *PNAS*, 119:e2121241119, 2022. *Co-first authorship [[URL](#)]
- 2021 Williams A., Pathmanathan J. S., **Stephens T. G.**, Su X., Chiles E. N., Conetta D., Putnam H. M., and Bhattacharya D. Multi-omic characterization of the thermal stress phenome in the stony coral *Montipora capitata*. *PeerJ*, 9:e12335, 2021. [[Preprint](#)] Featured by PeerJ Expert Curations: Molecular Ecology: [PeerJ Expert Curations](#), [PeerJ Blog](#) [[URL](#)]
- 2021 **Stephens T. G.**, Gabr A., Calatrevra V., Grossman A. R., and Bhattacharya D. Why is primary endosymbiosis so rare?. *New Phytologist*, 231:1693-1699, 2021. [[URL](#)]
- 2021 Gonzalez-Pech, R. A., **Stephens T. G.**, Chen Y., Mohamed A. R., Cheng Y., Shah S., Dougan K. E., Fortuin M. D. A., Lagorce R., Burt D. W., Bhattacharya D., Ragan M. A., and Chan C. X. Comparison of 15 dinoflagellate genomes reveals extensive sequence and structural divergence in family Symbiodiniaceae and genus *Symbiodinium*. *BMC Biology*, 0.842361111, 2021. [[Preprint](#)] [[URL](#)]
- 2020 **Stephens T. G.**, Gonzalez-Pech R. A., Cheng Y., Mohamed A. R., Burt D. W., Bhattacharya D., Ragan M. A., and Chan C. X. Genomes of the dinoflagellate *Polarella glacialis* encode tandemly repeated single-exon genes with adaptive functions. *BMC Biology*, 18:56, 2020. [[Preprint](#)] Featured by [IMB News](#) [[URL](#)]
- 2019 Chen Y., Gonzalez-Pech R. A., **Stephens T. G.**, Bhattacharya D., and Chan C. X. Evidence that inconsistent gene prediction can mislead analysis of dinoflagellate genomes. *Journal of Phycology*, 56:6-10, 2019. [[Preprint](#)] [[URL](#)]
- 2018 **Stephens T. G.**, Ragan M. A., Bhattacharya D., and Chan C. X. Core genes in diverse dinoflagellate lineages include a wealth of conserved dark genes with unknown functions. *Scientific Reports*, 8:17175, 2018. [[URL](#)]
- 2018 Liu H., **Stephens T. G.**, Gonzalez-Pech R. A., Beltran V. H., Lapeyre B., Bongaerts P., Cooke I., Aranda M., Bourne D. G., Foret S., Miller D. J., van Oppen M. J. H., Voolstra C. R., Ragan M.A., and Chan C.X. *Symbiodinium* genomes reveal adaptive evolution of functions related to coral-dinoflagellate symbiosis. *Communications Biology*, 0.107638889, 2018. [[Preprint](#)] Featured by multiple outlets: [IMB News](#), [GBRF](#), [Video Feature](#), [IMB 2018 year in review](#) [[URL](#)]

Selected Grants

- 2024 New Jersey CSIT Clean Tech R&D Voucher Pilot Program Round 3, \$40,000 USD
Funds awarded to our startup OceanOmics to utilize government facilities to aid development of our coral health monitoring toolkit.
Principal Investigators: **Timothy Stephens** (Rutgers University), New Jersey Commission on Science, Innovation and Technology
- 2024 New Jersey CSIT Clean Tech Seed Grant Program Round 3, \$75,000 USD
Funds awarded to our startup OceanOmics to continue development of our coral health monitoring toolkit.
Principal Investigators: **Timothy Stephens** (Rutgers University), New Jersey Commission on Science, Innovation and Technology

2023 Rutgers TechAdvance Fund, \$74,842 USD
Funds to continue development of our coral health monitoring toolkit.
Principal Investigators: Debashish Bhattacharya (Rutgers University), Rutgers University

Selected Teaching experience

2020 — present Co-lecturer
Rutgers University
Design and conducted lectures on de novo next-generation genome and transcriptome sequencing, metabolomics, proteomics, metagenomics, phylogenetics, and phylogenomics in a joint undergraduate and postgraduate course titled “Fundamentals of Microbial Genomics”. Duties include designing and marking student’ exam questions and oral presentations.

Selected Talks

- 2024 **Stephens T. G.**, Kulczyk A. W., and Bhattacharya D. Cosmopolitan gene families with known functions are hotspots for the evolution of novel genes in stony corals. *3rd Joint Congress on Evolutionary Biology*. 26-30 July 2024, Montreal, Canada.
- 2023 **Stephens T. G.**, Etten J. V., Benites L. F., Mcdermott T., and Bhattacharya D. Cyanidiophyceae: The extremophilic red algae that underpin hot spring microbial communities in Yellowstone National Park. *UQ Marine Ecogenomics Symposium 2023*. 13-14 December 2023, University of Queensland, Australia.
- 2023 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Exploring the origin and evolution of primary plastids using *Paulinella* as a model system. *International Society of Endocytobiology conference 2023*. 10-14th September 2023, Field Museum, Chicago.
- 2023 **Stephens T. G.**, Chille E., Strand E. L., Putnam H. M., and Bhattacharya D. Multi-omics investigation of coral resilience. *Invited talk at Carnegie Plant Biology Seminar, August 2023*. 25th August 2023, Carnegie Department of Plant Biology, Stanford University, USA.
- 2023 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Exploring the origin and evolution of primary plastids using *Paulinella* as a model system. *New Phytologist Next Generation Scientists Conference 2023*. 3rd July 2023, National University of Singapore, Singapore.
- 2022 **Stephens T. G.**, Strand E. L., Putnam H. M., and Bhattacharya D. Differences in ploidy and the prevalence of clonal propagation between *Montipora capitata* and *Pocillopora acuta* from Kane’ohe Bay, Hawai’i. *15th International Coral Reef Symposium*. 3-8th July 2022, Bremen, Germany.

Selected Media coverage

- 2024 Our Sargassum Biofuel project was featured by [Rutgers News](#)
- 2024 Our paper [Viruses associated with extremophilic red algal mats reveal signatures of early thermal adaptation](#) was featured by [Science](#)
- 2022 Article by [Rutgers Research](#) on our short film [The Coral Holobiont Response to Climate Change](#) which won Best Trailer in the Kiez Berlin Film Festival.
- 2020 Our paper [Amoeba Genome Reveals Dominant Host Contribution to Plastid Endosymbiosis](#) covered by [Rutgers Today](#)

Selected Service

2023 — present Associate Editor: Symbiotic and Parasitic Protists section, [Frontiers in Protistology](#)

Patents

- 2023 Provisional Patent “Protein Markers of Coral Stress” (5431.1025-000:2023-098)
- 2023 Provisional Patent “Coral Rapid Strip Test” (5431.1024-000:2023-097)